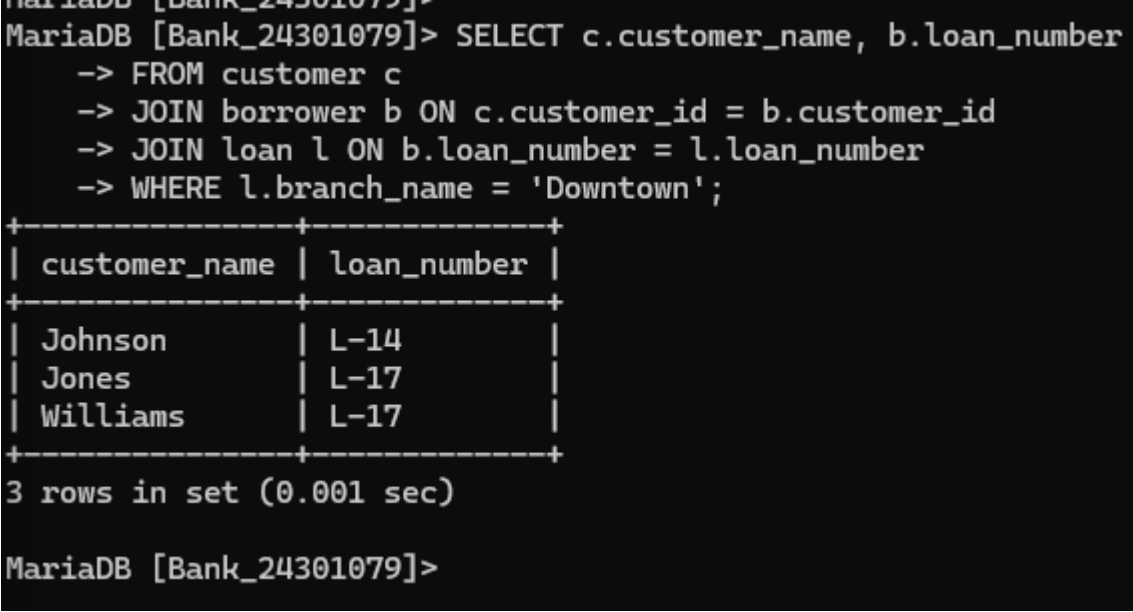


CSE370 : Database Systems

Assignment 03 | Spring 2026

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No 1 Query (as Plain Text)	<pre>SELECT c.customer_name, b.loan_number -> FROM customer c -> JOIN borrower b ON c.customer_id = b.customer_id -> JOIN loan l ON b.loan_number = l.loan_number -> WHERE l.branch_name = 'Downtown';</pre>
No 1 SS (of Query & Output in Shell)	
No 2 Query (as Plain Text)	<pre>SELECT DISTINCT c.customer_name -> FROM customer c</pre>

	-> JOIN borrower b ON c.customer_id = b.customer_id;
No 2 SS (of Query & Output in Shell)	<pre> MariaDB [Bank_24301079]> SELECT DISTINCT c.customer_name -> FROM customer c -> JOIN borrower b ON c.customer_id = b.customer_id; +-----+ customer_name +-----+ Jones Smith Hayes Curry Williams Adams Johnson +-----+ 7 rows in set (0.001 sec) </pre>
No 3 Query (as Plain Text)	<pre> SELECT c.customer_name -> FROM customer c -> WHERE c.customer_id NOT IN (-> SELECT customer_id FROM borrower ->); </pre>

No 3 SS
(of Query & Output
in Shell)

```
MariaDB [Bank_24301079]> SELECT c.customer_name
-> FROM customer c
-> WHERE c.customer_id NOT IN (
->     SELECT customer_id FROM borrower
-> );
```

customer_name
Lindsay
Turner
Glenn
Brooks
Green

```
5 rows in set (0.001 sec)
```

No 4 Query
(as Plain Text)

```
SELECT c1.customer_name AS Customer1, c2.customer_name AS Customer2, c1.customer_city AS City
FROM customer c1
JOIN customer c2
ON c1.customer_city = c2.customer_city
AND c1.customer_id < c2.customer_id;
```

No 4 SS
(of Query & Output
in Shell)

```
MariaDB [Bank_24301079]> SELECT c1.customer_name AS Customer1, c2.customer_name AS Customer2, c1.customer_city AS City
-> FROM customer c1
-> JOIN customer c2
-> ON c1.customer_city = c2.customer_city
-> AND c1.customer_id < c2.customer_id;

+-----+-----+-----+
| Customer1 | Customer2 | City |
+-----+-----+-----+
| Jones     | Hayes     | Harrison |
| Smith     | Curry     | Rye      |
| Lindsay   | Adams     | Pittsfield |
| Turner    | Green     | Stamford |
+-----+-----+-----+
4 rows in set (0.001 sec)
```

No 5 Query
(as Plain Text)

```
SELECT branch_name, ROUND(SUM(balance * 0.04), 2) AS Total_Interest
FROM account
GROUP BY branch_name;
```

No 5 SS
(of Query & Output
in Shell)

```
MariaDB [Bank_24301079]> SELECT branch_name, ROUND(SUM(balance * 0.04), 2) AS Total_Interest
-> FROM account
-> GROUP BY branch_name;
```

branch_name	Total_Interest
Brighton	66.00
Downtown	20.00
Mianus	28.00
Perryridge	16.00
Redwood	28.00
Round Hill	14.00

```
6 rows in set (0.000 sec)
```

No 6 Query
(as Plain Text)

```
SELECT a.account_number
-> FROM account a
-> JOIN branch b ON a.branch_name = b.branch_name
-> WHERE a.balance = (
->   SELECT MAX(a2.balance)
->   FROM account a2
->   JOIN branch b2 ON a2.branch_name = b2.branch_name
->   WHERE b2.branch_city = b.branch_city
-> );
```

No 6 SS
(of Query & Output
in Shell)

```
MariaDB [Bank_24301079]> SELECT a.account_number
-> FROM account a
-> JOIN branch b ON a.branch_name = b.branch_name
-> WHERE a.balance = (
->     SELECT MAX(a2.balance)
->     FROM account a2
->     JOIN branch b2 ON a2.branch_name = b2.branch_name
->     WHERE b2.branch_city = b.branch_city
-> );
```

account_number
A-201
A-215
A-222

```
3 rows in set (0.001 sec)
```

No 7 Query
(as Plain Text)

```
SELECT * FROM (
  SELECT l.loan_number, l.amount, c.customer_name
  from loan l
  JOIN borrower b ON l.loan_number = b.loan_number
  JOIN customer c ON b.customer_id = c.customer_id
  order by l.amount DESC
  limit 5
) AS TopLoans
ORDER BY amount ASC, loan_number DESC;
```

No 7 SS
(of Query & Output
in Shell)

```
MariaDB [Bank_24301079]> SELECT * FROM (
->     SELECT l.loan_number, l.amount, c.customer_name
->     from loan l
->     JOIN borrower b ON l.loan_number = b.loan_number
->     JOIN customer c ON b.customer_id = c.customer_id
->     order by l.amount DESC
->     limit 5
-> ) AS TopLoans
-> ORDER BY amount ASC, loan_number DESC;
+-----+-----+-----+
| loan_number | amount | customer_name |
+-----+-----+-----+
| L-17       | 1000   | Jones        |
| L-16       | 1300   | Adams        |
| L-15       | 1500   | Hayes        |
| L-14       | 1500   | Johnson      |
| L-23       | 2000   | Smith        |
+-----+-----+-----+
5 rows in set (0.000 sec)
```

No 8 Query
(as Plain Text)

```
SELECT DISTINCT c.customer_name
-> FROM customer c
-> JOIN depositor d ON c.customer_id = d.customer_id
-> JOIN account a ON d.account_number = a.account_number
-> JOIN borrower b ON c.customer_id = b.customer_id
-> JOIN loan l ON b.loan_number = l.loan_number
-> WHERE a.branch_name = 'Perryridge'
```

	-> AND l.branch_name = 'Perryridge';
No 8 SS (of Query & Output in Shell)	<pre> MariaDB [Bank_24301079]> SELECT DISTINCT c.customer_name -> FROM customer c -> JOIN depositor d ON c.customer_id = d.customer_id -> JOIN account a ON d.account_number = a.account_number -> JOIN borrower b ON c.customer_id = b.customer_id -> JOIN loan l ON b.loan_number = l.loan_number -> WHERE a.branch_name = 'Perryridge' -> AND l.branch_name = 'Perryridge'; +-----+ customer_name +-----+ Hayes +-----+ 1 row in set (0.000 sec) </pre>
No 9 Query (as Plain Text)	<pre> SELECT c.customer_name, SUM(l.amount) AS total_loan FROM customer c JOIN borrower b ON c.customer_id = b.customer_id JOIN loan l ON b.loan_number = l.loan_number GROUP BY c.customer_name HAVING COUNT(l.loan_number) >= 2; </pre>

No 9 SS
(of Query & Output
in Shell)

```
MariaDB [Bank_24301079]> SELECT c.customer_name, SUM(l.amount) AS total_loan
-> FROM customer c
-> JOIN borrower b ON c.customer_id = b.customer_id
-> JOIN loan l ON b.loan_number = l.loan_number
-> GROUP BY c.customer_name
-> HAVING COUNT(l.loan_number) >= 2;

+-----+-----+
| customer_name | total_loan |
+-----+-----+
| Smith         |         2900 |
+-----+-----+
1 row in set (0.001 sec)
```

No 10 Query
(as Plain Text)

```
SELECT d.customer_id, SUM(a.balance) AS total_balance
FROM depositor d
JOIN account a ON d.account_number = a.account_number
GROUP BY d.customer_id
HAVING COUNT(d.account_number) > 1;
```

No 10 SS
(of Query & Output
in Shell)

```
MariaDB [Bank_24301079]> SELECT d.customer_id, SUM(a.balance) AS total_balance  
-> FROM depositor d  
-> JOIN account a ON d.account_number = a.account_number  
-> GROUP BY d.customer_id  
-> HAVING COUNT(d.account_number) > 1;  
  
+-----+-----+  
| customer_id | total_balance |  
+-----+-----+  
| C-226      |          1400 |  
+-----+-----+  
1 row in set (0.000 sec)
```